Vibe Coding

Jiugeng Sun & Erika Soans

What is ‘vibe coding’?

Describing what you want in plain language



Andrej Karpathy

I like to train deep neural nets on large datasets 🧠🤖💥

2024 -

I create educational videos on AI on my YouTube channel. The videos come in two parallel tracks: a technical track and a general audience track.

1. Technical track: Follow the Zero to Hero playlist. General audience track:
2. Deep Dive into LLMs like ChatGPT is on under-the hood fundamentals of LLMs.
3. How I use LLMs is a more practical guide to examples of use in my own life.
4. Intro to Large Language Models is a third, parallel, video from a longer time ago.

For all the latest, I spend most of my time on 𝕏/Twitter or GitHub.

Andrej Karpathy ✓
@karpathy

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

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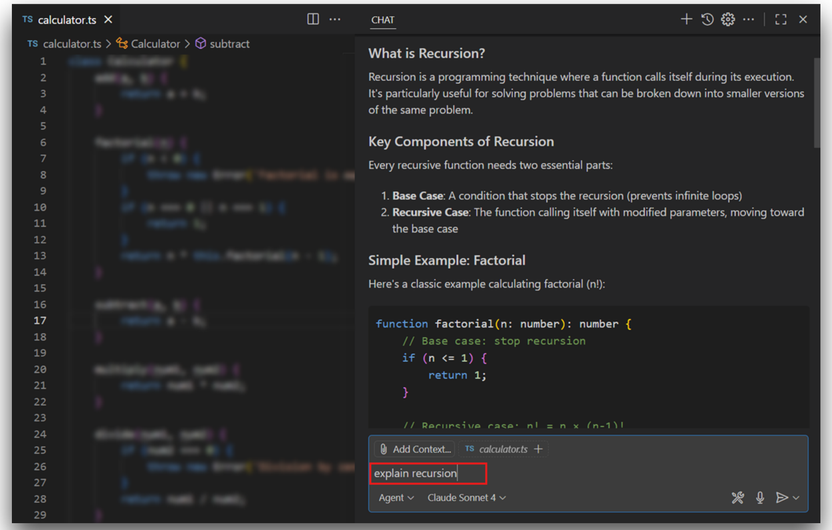
You're probably already doing this..

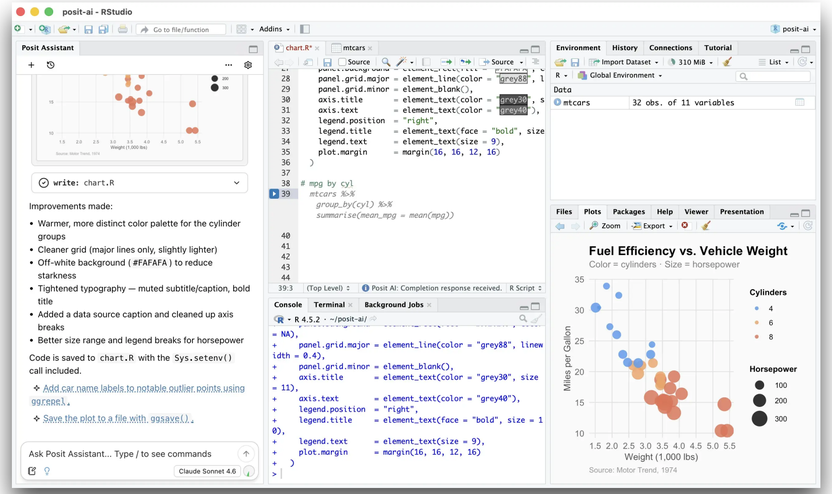
Tool	What it could see/do	Feels like to you
Browser chat (claude.ai, ChatGPT)	Only what you paste	Copy-paste
Inline completion (Copilot)	The current programming file	Autocomplete++
IDE chat/agent (Posit, Cursor, VS Code + Claude or Codex extension)	The files and live programming session	pair programmer
Terminal agent (Claude Code, Codex CLI)	Your whole project folder; can run commands	Task delegation

IDE chat & agent

The agent that lives in your editor

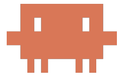
- Cursor
  - a VS Code Studio fork, with the agent built in from the start
- VS Code Studio
  - Claude, Codex, Gemini, and Copilot all have extensions here
- R Studio Posit AI
  - Posit Assistant reads your live R session, loaded packages, environment variables, and data frames …

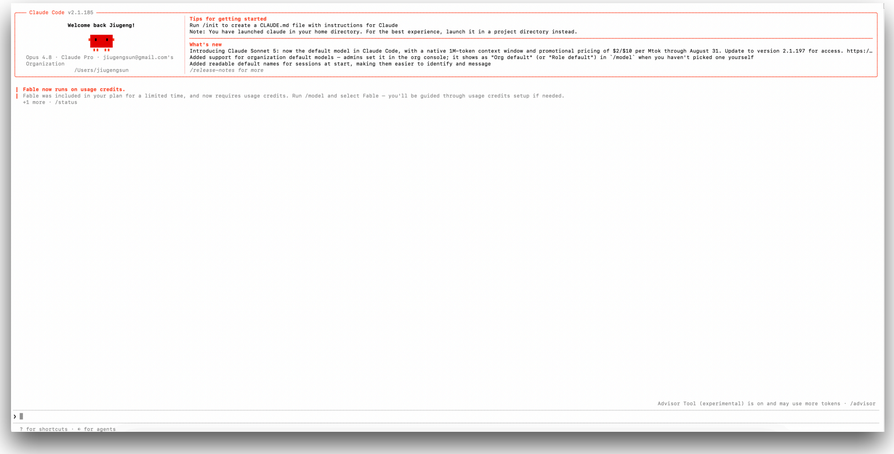




Terminal chat & agent

The agent that lives in your command line

- Claude Code  Codex CLI  Gemini CLI 
  - **Editor & language-agnostic**:
    - it doesn't care if you are an R person or a Python person who uses either R Studio or other IDEs
  - **Budget saving**:
    - it comes with the subscription you already have (Claude Pro…); Cursor or Posit asks for subscriptions
  - **Remote connection**:
    - with SSH it could run diagnosis on cloud clusters



Let’s be aware…



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Be careful!

ETH guidelines for safe use of AI

- ETH encourages use of AI while being mindful of the risks involved - ‘exploit opportunities, minimise risk’.
- Be careful with sharing sensitive data onto AI platforms/softwares.
- Breach of sensitive data (eg. survey info) should be reported to the ETH data protection officer.
- AI systems are classified as third-party service. ETH is **still working on** the list of approved AI systems.

Be careful!

Prudence and Care

- ‘Employee may use AI in their day-to-day work’
- Dos
  - Write meaningful prompts
  - Critically review the results of AI
  - Disable the history function
  - Be sure to consider copyright using AI-generated content
- Donts
  - Don’t share confidential information with AI systems
  - Don’t share personal data
  - Don’t blindly rely on AI results - lack of coherence; semantic errors; repetition; hallucination
  - Consider ethical guidelines

https://ethz.ch/staffnet/en/news-and-events/internal-news/archive/2025/09/new-guidelines-for-the-safe-use-of-ai-at-eth-zurich.html

Let’s get geeky…

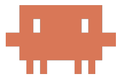
INSTALL.SH

```
#!/bin/bash

pip install "$1" &
easy_install "$1" &
brew install "$1" &
npm install "$1" &
yum install "$1" & dnf install "$1" &
docker run "$1" &
pkg install "$1" &
apt-get install "$1" &
sudo apt-get install "$1" &
steamcmd +app_update "$1" validate &
git clone https://github.com/"$1"/"$1" &
cd "$1";./configure;make;make install &
curl "$1" | bash &
```

Let’s get hands-on exercise!

Pick your choice

- Claude Code 
  - https://code.claude.com/docs/en/quickstart
- Gemini CLI 
  - https://geminicli.com/docs/get-started/installation/
- Codex CLI 
  - https://github.com/openai/codex#quickstart

Claude code as an example

Three commands, one login, and the docs for everything else

- Install the coding agent through **command line**

```
curl -fsSL https://claude.ai/install.sh | bash          # macOS / Linux
```

```
irm https://claude.ai/install.ps1 | iex                 # Windows PowerShell
```

- Check it's there `claude --version`
- you need a paid account: Pro, Max, Team or Enterprise. The free Claude.ai plan doesn't include Claude Code.
- code.claude.com/docs

First session

type `claude` and describe what you want

Find the an example CSV file here: smb://ites-em.ethz.ch/data/EM_share/ErikaS/AI Session/field_observations.csv

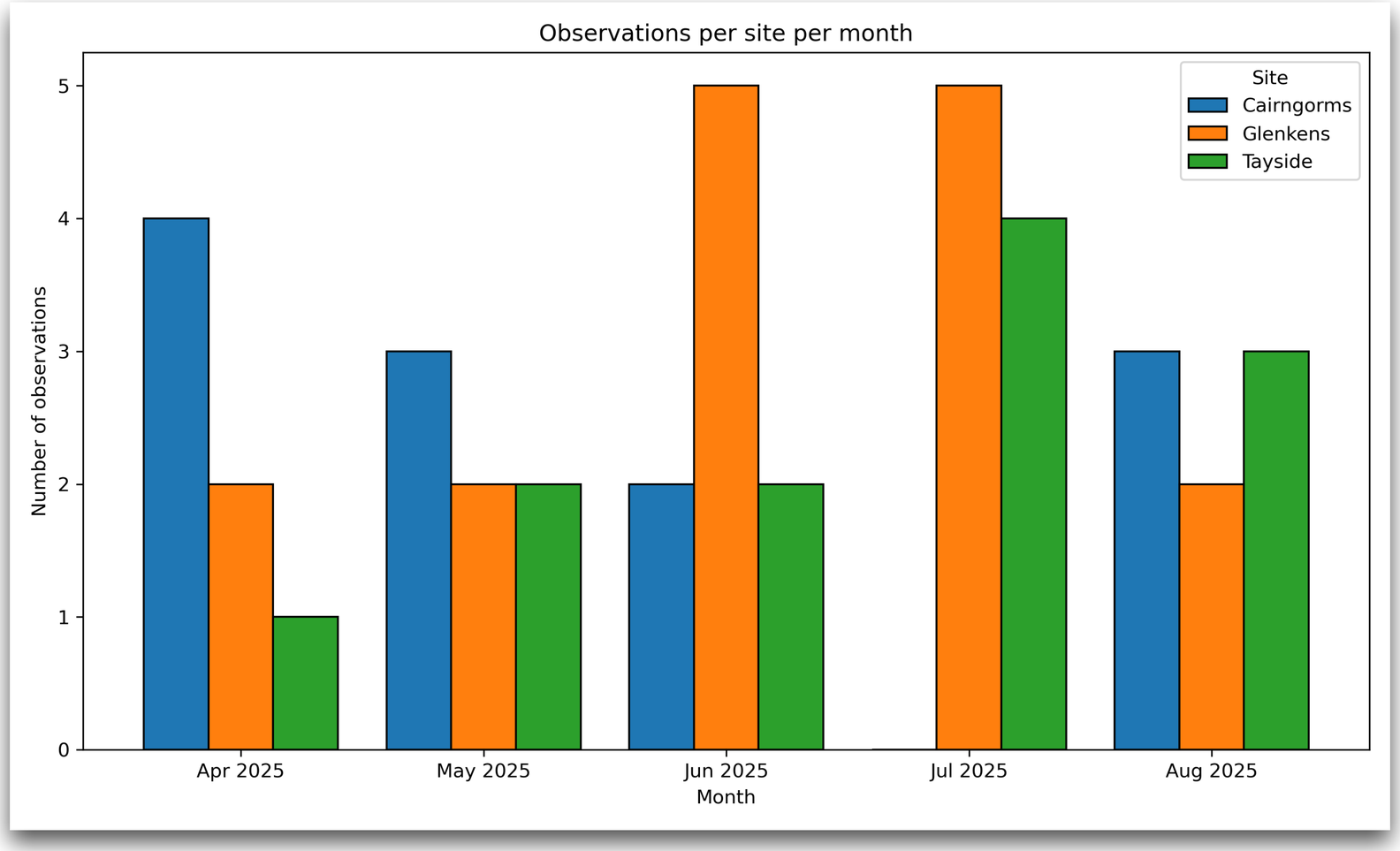
- Start it, and log in

```
cd your-project-folder
claude
```

- Prompting the text with:
  - Look at this CSV and tell me what's wrong with it
  - Write an **X (Your preferred programming language)** script that cleans it, and explain your choices
  - Now plot observations per site per month

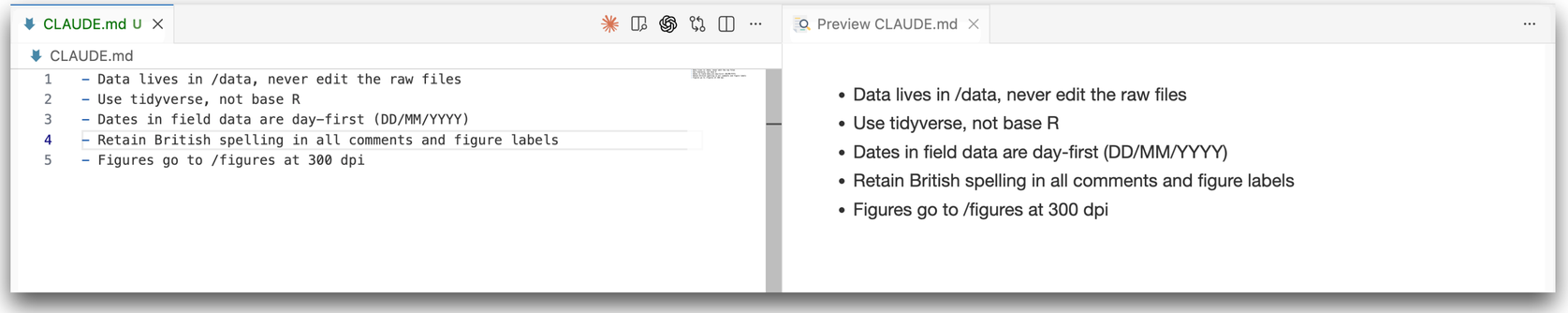
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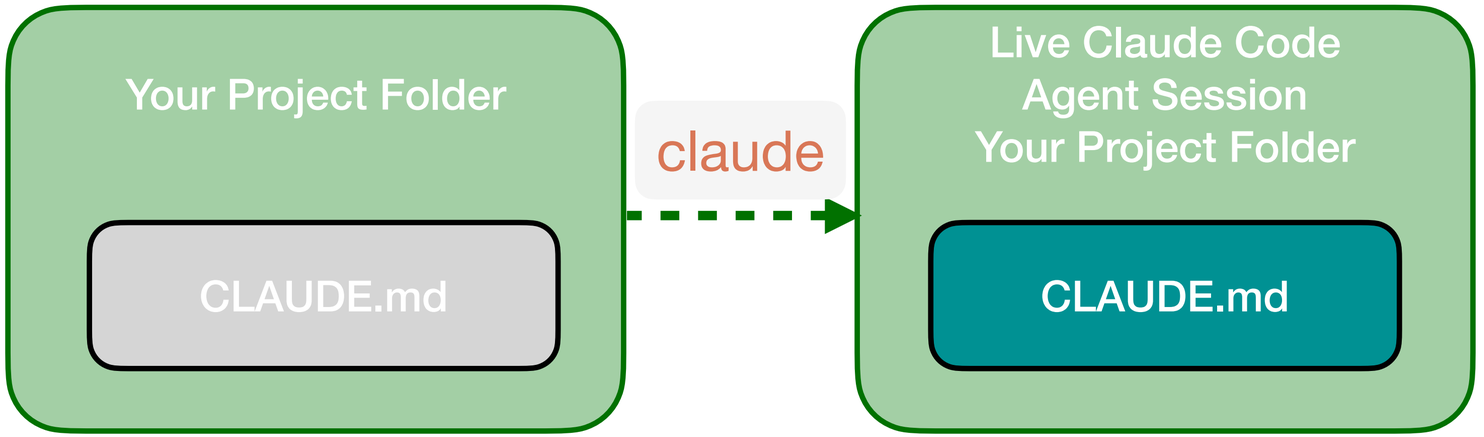


Stop repeating yourself

Write your project's conventions down once, in a file it reads every time



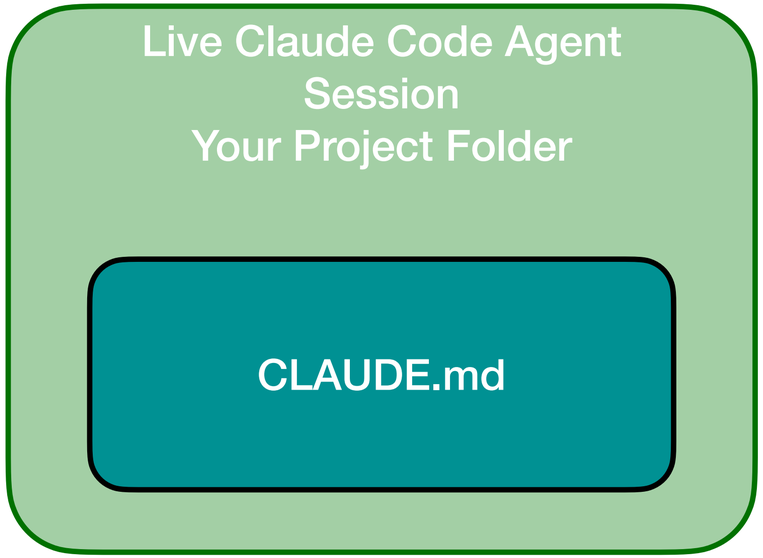
- Your agent can have more context
- Crate `CLAUDE.md` in the root
- `CLAUDE.md` is a special file that Claude reads at the start of every conversation.
- Put in conventions, not instructions here:
  where data lives, which packages, how dates are formatted



Skill

Instruct your agent to gain more “characteristics”

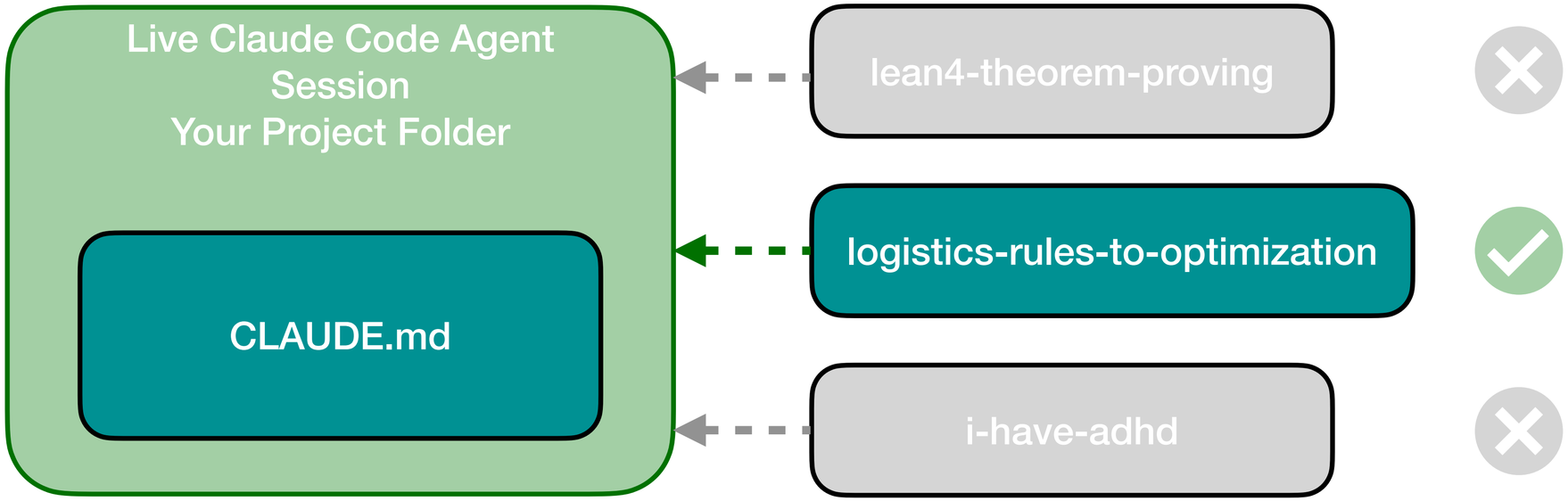
- Usually is a folder with `SKILL.md` inside
- You activate the skill only when you feel like you need it



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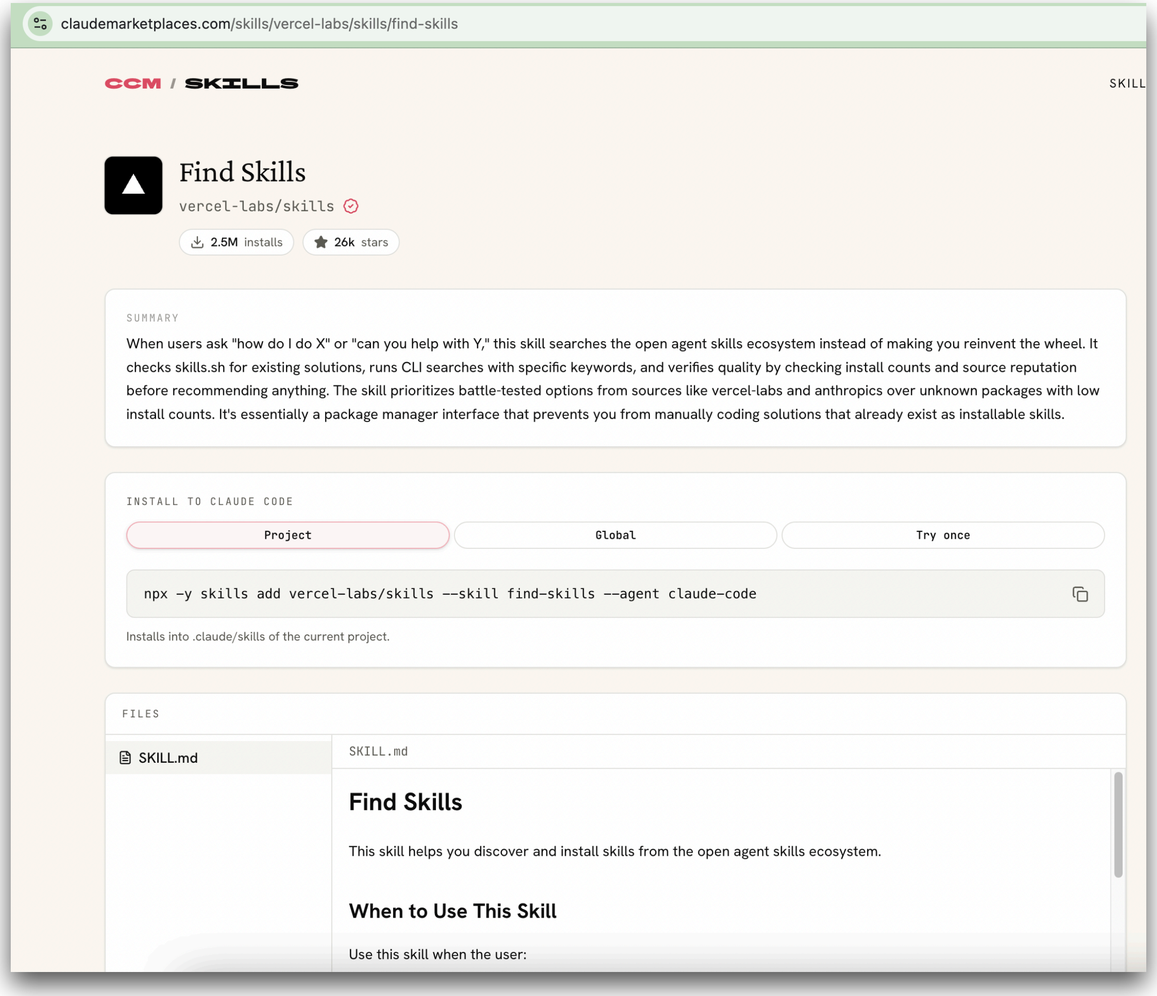
Skill

How do I find skills?

- So far there's no app store
  You need to look for the skills online
- Anthropic publishes some:

  anthropics/skills

  anthropics/claude-plugins-official
- There are even skills to help you find a skill



Skill

An example



ADHD-friendly outputs. No ADHD diagnosis needed!



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Install

🔗 Installation Instructions

What it does

A skill for your coding assistant that stops it from burying the answer. Action first. Steps numbered. No "Hope this helps!"

What changes

Before

> Great question! Let me think about this. Your auth flow has a few moving pieces: the middleware, the token verification, and the cookie handling. Looking at `src/auth.ts`, the `verifyToken` function (around lines 42-58) seems to be using an older `jsonwebtoken` API. One approach would be to update the package and rewrite that function. After making the change, you'd want to run the auth tests to confirm nothing breaks. By the way, you might also want to look at your dependency versions overall. Hope this helps! Let me know if you want to dig deeper.

After

> Run `npm install jsonwebtoken@latest`, then edit `src/auth.ts:42`.
>
> 1. Open `src/auth.ts`
> 2. Replace `verifyToken` (lines 42–58) with the snippet below
> 3. Run `npm test -- auth.spec.ts`
>
> Next: paste the first failing line if any test fails.

The rules

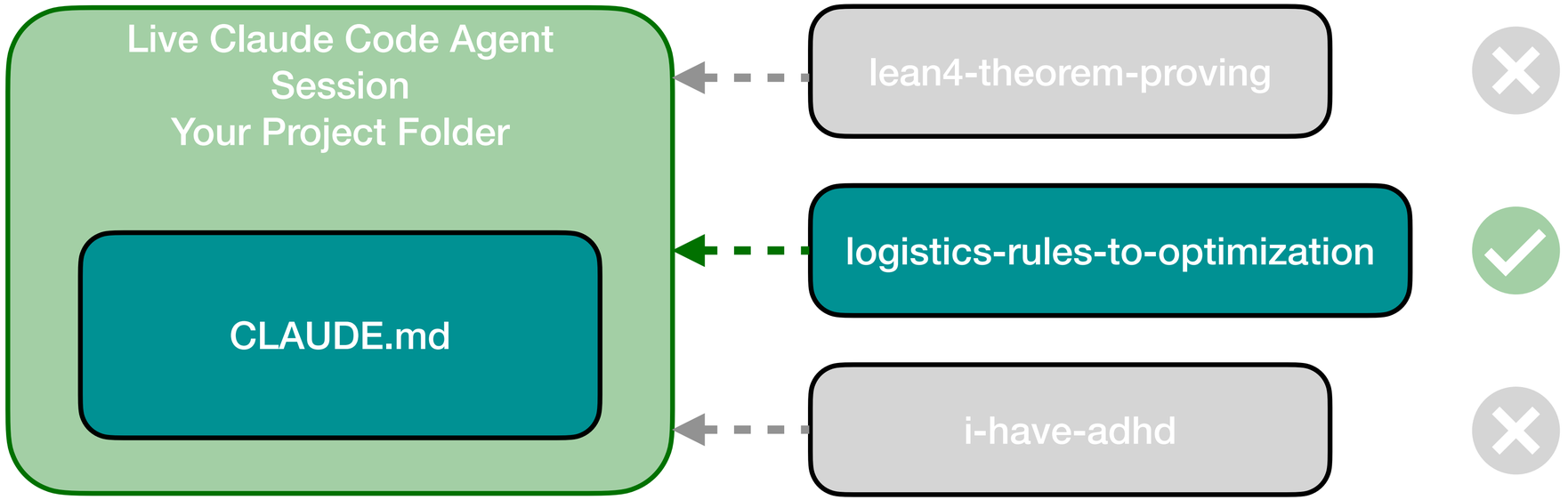
10 rules. Full text in SKILL.md.

1. Lead with the next action.
2. Number multi-step tasks.
3. End with one concrete next step.
4. Suppress tangents.
5. Restate state every turn.
6. Specific time estimates (minutes, not "a bit").
7. Make wins visible.
8. Matter-of-fact errors.
9. Cap lists at 5 items.
10. No preamble. No recap. No closers.

MCP: plugging in your other resources

Zotero, your database, Google Drive — without leaving the terminal

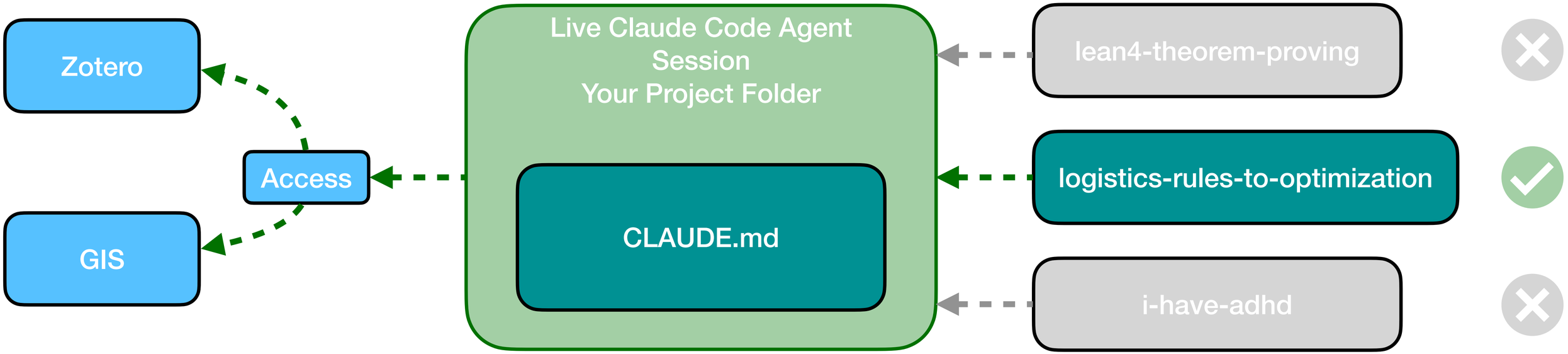
- MCP (Model Context Protocol): a standard way for an agent to reach systems outside your folder
- a Skill is instructions, an MCP server is a connection. `claude mcp add`



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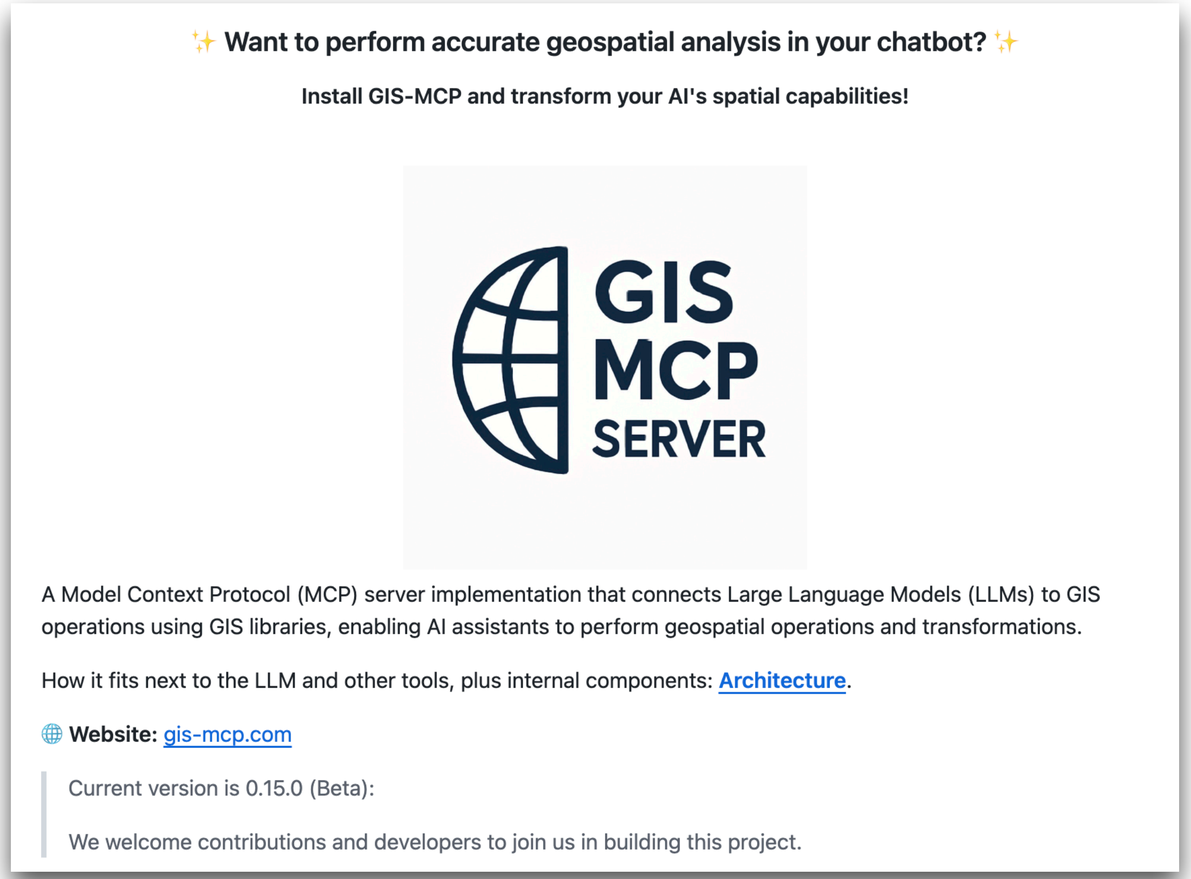
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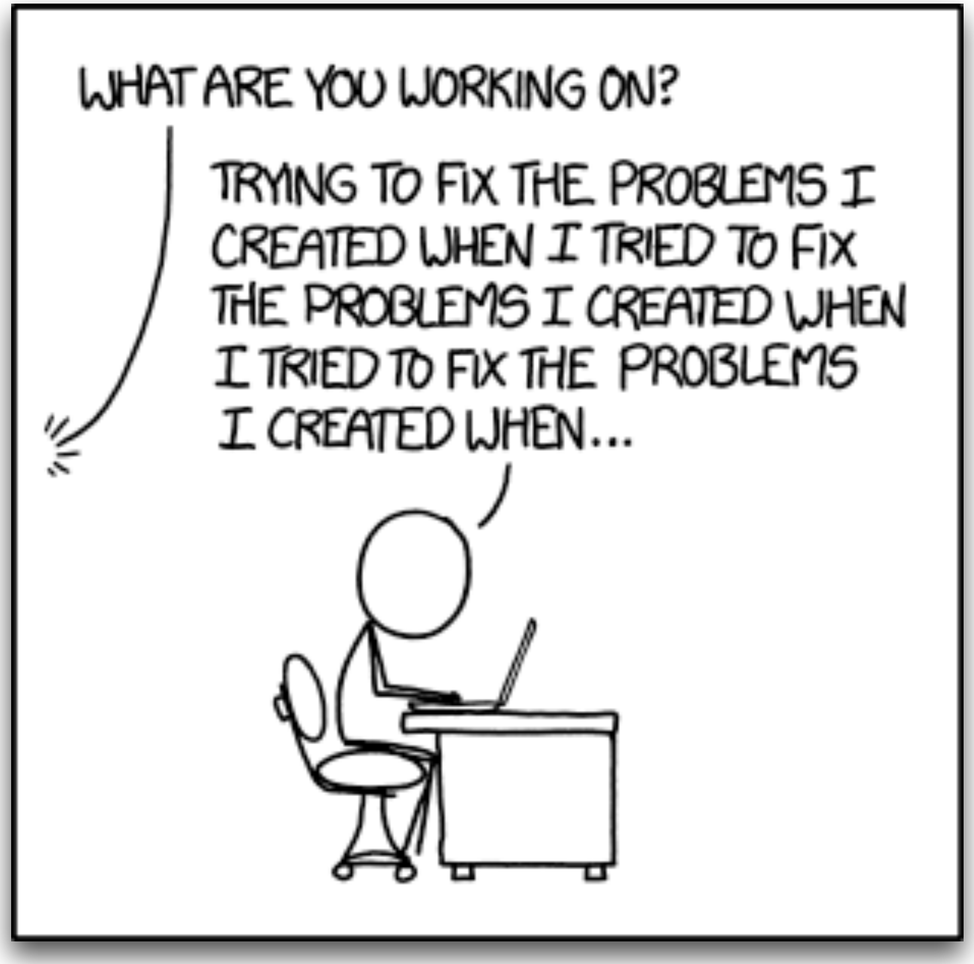
MCP (Model Context Protocol)

An example

- **gis-mcp**: connects LLMs/AI coding agent to GIS operations using standard geospatial libraries.



But…


WHAT ARE YOU WORKING ON?
TRYING TO FIX THE PROBLEMS I CREATED WHEN I TRIED TO FIX THE PROBLEMS I CREATED WHEN I TRIED TO FIX THE PROBLEMS I CREATED WHEN...

What NOT to do

The bit that's different when it can edit your files

- Don't accept a big refactor on trust
  > ask for a plan first, agree it or break it into small tasks, then let it/them run
- Don't accept the permission prompt immediately; it usually asks with a reason
- Don't point it at your only copy of the raw data
- Don't run one endless session
  > Quality drifts as context fills. New task, new session.
- Better not work without Git
- Avoid over-engineering: fix the bug by key-word searching for example

Best practice

You can be a master of vibe coding if you know all of these

- Use Git control: `git init` your analysis folder
- Start to write `CLAUDE.md` for your project in your analysis folder
- Remember what you did with code agent
  - `/export <file>` writes the current conversation out as a transcript
  - Claude Code automatically stores every session as a plaintext `.jsonl` under `~/.claude/projects/` ; auto-deleted after 30 days by default raise `cleanupPeriodDays` in `~/.claude/settings.json` to keep them longer
- https://code.claude.com/docs/en/best-practices

Thank you for listening :)

